

HEALTHCONNECT EXPERIENCE LAB

Project Title: Healthconnect Clinic Initial Analysis Document

Program: AnalystLab Africa Data Analytics Internship

Week 4

Project Background

HealthConnect Clinic is a healthcare provider that manages appointment-based services. The clinic is experiencing operational and patient-support challenges, including missed appointments, inefficient utilisation of appointment slots, difficulty understanding factors associated with appointment no-shows, and repetitive appointment-related enquiries.

The HealthConnect Experience Lab aims to explore how data can support better decision-making, improve appointment management, reduce missed appointments and enhance the patient support experience.

As part of the Data Analytics track, this initial analysis focuses on understanding the HealthConnect appointment dataset, assessing its quality, identifying variables relevant to appointment attendance, defining meaningful business questions and proposing key performance indicators for subsequent analysis.

Central Project Question: How can HealthConnect Clinic use data to reduce missed appointments and improve the patient support experience?

Data Analytics Track Objective

The objective of the Data Analytics track is to investigate appointment attendance and no-show patterns using the available appointment data.

The analysis will focus on identifying:

- Patterns associated with appointment no-shows.
- Patient and appointment characteristics that may influence attendance.
- The potential relationship between reminders and appointment outcomes.
- The relationship between previous no-shows and future attendance.
- Operational factors that may require further investigation.
- KPIs that can support future decision-making.

This Week 4 document establishes the analytical foundation for subsequent project stages rather than presenting a completed dashboard, predictive model or final solution.

Dataset Overview

Description: The HealthConnect Appointment Dataset is a dataset containing appointment records for HealthConnect Clinic.

The dataset contains:

- 5,000 appointment records
- 18 variables

The variables capture information across several areas:

Patient Demographics

- Gender
- Age
- Age Group

Appointment Information

- Appointment Type
- Booking Date
- Appointment Date
- Appointment Day
- Appointment Time

Booking Behaviour

- Booking Lead Days

Patient History

- Previous Appointments
- Previous No-Shows

Reminder Information

- Reminder Sent
- Reminder Channel

Operational Factors

- Distance to Clinic
- Waiting Time Minutes

Appointment Outcome

- Appointment Outcome

The dataset therefore provides information that can be used to examine relationships between patient characteristics, appointment characteristics, previous behaviour, reminder practices and appointment outcomes.

Data Dictionary Summary

S/N	Variable	Data Type	Analytical Relevance
1.	Appointment ID	Text	Uniquely identifies appointments
2.	Patient ID	Text	Identifies patients across appointment
3.	Gender	Text	Supports demographic comparison
4.	Age	Integer	Supports age-based analysis
5.	Age Group	Text	Enables demographic segmentation
6.	Appointment Type	Text	Compares attendance across appointment types
7.	Booking Date	Date	Identifies when appointment was booked
8.	Appointment Date	Date	Identifies scheduled appointment date
9.	Appointment Day	Text	Enables day of week analysis
10.	Appointment Time	Text	Enables time period analysis
11.	Booking Lead Days	Integer	Measures time between booking and appointment
12.	Previous Appointment	Integer	Indicates historical appointment activity
13.	Previous No Shows	Integer	Indicates previous attendance behaviour
14.	Reminder Sent	Text	Indicates whether reminder was sent

15.	Reminder Channel	Text	Indicates reminder communication method
16.	Distance to Clinic (KM)	Decimal	Examines accessibility related patterns
17.	Waiting Time Minutes	Integer	Examines potential operational impact
18.	Appointment Outcome	Text	Primary outcome variable for attendance analysis

Initial Data Quality Assessment: An initial assessment was conducted to examine completeness, uniqueness, consistency, validity and data-type accuracy.

Dataset Size

The dataset contains:

- 5,000 rows
- 18 variables

This provides a sufficient starting point for descriptive and exploratory analysis within the scope of the project.

Missing-Value Assessment: The initial inspection identified missing values in two variables:

- Distance to clinic: 90 missing records (1.8 %)
- Waiting time (minutes): 60 missing records (1.2%)

Reminder Channel: The Reminder Channel field contains:

- Email
- SMS
- WhatsApp
- None

Data Quality Checks Requiring Further Validation: The following checks should be completed before the dataset is used for detailed analysis.

- Quality Check
- Appointment ID duplicates
- Appointment ID missing values
- Patient ID missing values
- Inconsistent text values
- Booking Date data type consistency
- Appointment Date data type consistency
- Invalid age values
- Negative waiting times
- Negative booking lead days
- Invalid appointment dates
- Date-Type Observation

The initial inspection indicates that some date fields may contain mixed representations. Booking Date and Appointment Date should therefore be standardised during the data-preparation stage.

The original project resources should be preserved, while any cleaned or transformed versions should be saved separately, in accordance with the project instructions.

Key Variables for No-Show Analysis: The following variables have been identified as particularly relevant to the central business problem.

Appointment Outcome: This is the primary outcome variable because it indicates whether an appointment was:

- Attended
- No-Show
- Cancelled

It will form the basis of the attendance and no-show analysis.

Previous No-Shows: Previous No-Shows can help determine whether past attendance behaviour is associated with future appointment outcomes.

Reminder Sent: This can be used to investigate whether appointments where reminders were sent demonstrate different attendance patterns from appointments without reminders.

Reminder Channel: This enables comparison of different reminder methods:

- SMS
- Email
- WhatsApp
- None

Booking Lead Days: This measures the time between booking and the scheduled appointment and may help identify whether longer booking periods are associated with increased no-show risk.

Appointment Type: Different appointment types may have different attendance patterns.

Appointment Day and Time: These variables allow attendance patterns to be examined across days of the week and appointment periods.

Age and Age Group: These variables allow demographic segments with potentially different attendance patterns to be identified.

Distance to Clinic: Distance may provide insight into whether accessibility is associated with appointment attendance.

Waiting Time: Waiting time may help investigate potential operational or patient-experience relationships with appointment outcomes.

Business Questions

The following questions will guide subsequent analysis.

1. What factors are most strongly associated with appointment no-shows?
2. Does sending appointment reminders relate to improved appointment attendance?
3. Are patients with previous no-shows more likely to miss future appointments?
4. Does booking lead time influence appointment attendance?
5. Which patient and appointment segments experience the highest no-show rates?

Potential segments include age group, appointment type, appointment day, appointment time, reminder status and previous no-show behaviour.

Proposed KPI

Five potential KPIs have been identified for subsequent stages of the project.

S/N	KPI	Related Business Questions	Relevance
1.	No-Show Rate	What factors are associated with missed appointments?	Measures the proportion of appointments missed
2.	Attendance Rate	How effectively are appointments being attended?	Measures successful appointment attendance
3.	Cancellation Rate	How frequent are appointments cancelled?	Measures appointment disruption
4.	Reminder Coverage Rate	Are reminders reaching scheduled patients	Evaluates reminder implementation

These KPIs are proposed for future analysis and are not presented as calculated results at this stage.

Initial Analysis Approach: The proposed analytical approach will be conducted in several stages.

1. **Data Understanding:** Review the dataset structure and data dictionary to understand the meaning, format and analytical relevance of each variable.
2. **Data Quality Assessment:** Assess; Missing values, Duplicate records, Data types, Invalid values, Inconsistent categories, Date consistency, logical relationships between variables.
3. **Data Preparation:** Create a separate working version of the dataset and, where necessary; Standardise data types. Correct inconsistent categorical values. Remove unnecessary whitespace. Resolve genuine duplicate records where appropriate. Handle missing values using an appropriate documented approach. Create analytical fields where required.

Exploratory Analysis: Explore appointment outcomes across; Age Group, Gender, Appointment Type, Appointment Day, Appointment Time, Booking Lead Days, Previous Appointments, Previous No-Shows, Reminder Sent, Reminder Channel, Distance to Clinic, Waiting Time.

KPI Development: Calculate the selected KPIs and compare them across relevant patient and appointment segments.

Visual Analysis: Use appropriate analytical visualisations to communicate patterns and relationships identified in the data. Power BI may subsequently be used to develop an interactive analytical dashboard.

Business Interpretation: Translate analytical findings into practical insights for HealthConnect Clinic. The objective will not simply be to identify which groups have higher no-show rates, but to understand where the clinic may have opportunities to improve appointment attendance and patient engagement.

Business Assumptions: The initial analysis is based on the following assumptions:

1. Appointment ID is intended to uniquely identify each appointment.
2. Patient ID may legitimately occur across multiple appointment records.
3. Appointment Outcome accurately represents the final status of each appointment.
4. None in Reminder Channel represents the absence of a reminder rather than missing information.
5. Previous No-Shows represent historical missed appointments associated with the patient.
6. Booking Lead Days represents the period between booking and the scheduled appointment.
7. Age and demographic information is assumed to be recorded consistently.
8. The dataset is fictional and anonymised and therefore does not represent actual clinical populations.
9. Observed relationships should not automatically be interpreted as causal relationships.

Limitations: Several limitations should be considered when interpreting the analysis.

1. Missing Data: Distance to Clinic contains 90 missing records and Waiting Time contains 60 missing records. These may limit analysis involving those variables.
2. Limited Context: The dataset may not contain every factor influencing appointment attendance, such as personal emergencies, transportation disruptions or changes in patient circumstances.
3. Observational Analysis: Relationships identified between variables and appointment outcomes do not necessarily establish causation.
4. Data Quality: Mixed date representations and any other inconsistencies identified during validation may require additional preprocessing before detailed analysis.
5. Patient-Level Interpretation: Patient identifiers are anonymised, limiting the ability to investigate individual patient circumstances.

Risks and Dependencies:

- Missing values: May affect some segment analyses
- Incorrect date formats: May affect time-based analysis
- Duplicate appointment records: Could distort appointment counts
- Inconsistent categories: Could produce misleading segment comparisons
- Limited contextual variables: May prevent explanation of some no-shows
- Synthetic data: Limits real-world generalisation
- Incorrect interpretation of relationships: Could result in inappropriate recommendations

Expected Analytical Outcome:

The analysis is expected to provide HealthConnect Clinic with a structured understanding of appointment attendance patterns and identify areas requiring deeper investigation.

The subsequent stages of the project should use the findings to:

- Identify major no-show patterns.
- Understand high-risk appointment segments.
- Evaluate reminder-related patterns.
- Monitor relevant KPIs.
- Support evidence-based operational recommendations.

Conclusion

The Week 4 analysis establishes the foundation for investigating missed appointments at HealthConnect Clinic.

The initial assessment confirms that the dataset contains 5,000 appointment records across 18 variables, covering patient demographics, appointment characteristics, booking behaviour, previous appointment history, reminders, operational factors and appointment outcomes.

Two variables currently contain identified missing values: Distance to Clinic with 90 missing records (1.8%) and Waiting Time Minutes with 60 missing records (1.2%). The None category in Reminder Channel is treated as a meaningful category because it represents appointments for which no reminder was sent.

The next stage will focus on completing data validation and preparation, conducting exploratory analysis, calculating the proposed KPIs and identifying patterns that can inform strategies for reducing missed appointments.